

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	October 2026
Team ID	6ab75a49da631abbf0cded44
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture

This deliverable adapts the supplied Technology Stack template to the ServiceNow standard-laptop procurement project. The architecture and technology entries below are grounded in the supplied project documentation; items not specified in that document are explicitly marked as “Not specified”.

Technical Architecture — Standard Laptop Procurement Automation



Request / Requested Item / Catalog Task status can be reviewed in ServiceNow

Architecture guidelines applied from the sample template

- Include the documented processes as application logic / technology blocks.
- Identify the platform and interfaces used by the workflow.
- Identify data, storage, external APIs and machine-learning components only where the project documentation supports them.
- Record infrastructure, security, scalability, availability and performance details without inventing undocumented implementation details.

Project Context

Problem Statement. The current IT procurement process lacks efficiency and automation, resulting in delays and manual overhead, particularly in handling standard laptop orders. Standard laptop requests often require configuration, but this step can be overlooked or delayed.

User Story. As a member of the IT procurement team, I want a streamlined process for ordering standard laptops so that tasks are automatically generated and assigned to the hardware team for configuration. This is intended to reduce user wait time and manual intervention.

Project Objective. Implement an automated workflow using Flow Designer to facilitate procurement and configuration of standard laptops, with the goals of timely configuration, reduced manual intervention and errors, improved task allocation, and enhanced IT procurement efficiency.

The project documentation explicitly identifies Flow Designer, Service Catalog, Catalog Task creation, the Hardware assignment group, the Standard Laptop catalog item, approval status, and the System user execution context.

Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	How the user interacts with the application: the Service Catalog / Standard Laptop catalog item is used to place the request.	ServiceNow Service Catalog
2	Application Logic-1	Automation logic that triggers after the relevant Service Catalog event and creates a Catalog Task.	ServiceNow Flow Designer
3	Application Logic-2	Catalog item process configuration connects the Standard Laptop service item to the created flow.	ServiceNow Maintain Items / Process engine
4	Application Logic-3	Task creation logic sets short description, description, Assignment group and Approval values.	Create Catalog Task action
5	Database	Data store for requests, requested items and catalog tasks is not separately specified in the source project document.	Not specified
6	Cloud Database	No separate cloud database service is identified in the source project document.	Not specified
7	File Storage	No file-storage requirement is identified in the source project document.	Not specified
8	External API-1	No external API is identified in the documented workflow.	Not specified / None documented
9	External API-2	No second external API is identified in the documented workflow.	Not specified / None documented
10	Machine Learning Model	No machine-learning model is described or required in the documented workflow.	Not applicable / None documented
11	Infrastructure (Server / Cloud)	The project is implemented through the ServiceNow application environment; specific server/cloud configuration is not documented.	ServiceNow platform; configuration not specified

Table 2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No open-source framework is identified in the supplied project documentation.	Not specified
2	Security Implementations	The documented workflow uses approval status and specifies "System user" as the Flow execution user. Detailed encryption, IAM, firewall or OWASP controls are not documented.	ServiceNow approval; System user
3	Scalable Architecture	The source document does not provide a scalability design, request-rate target, tiering model or microservices architecture.	Not specified
4	Availability	The source document does not specify load balancers, distributed servers, failover or availability targets.	Not specified
5	Performance	The project objective is to reduce manual overhead and user wait time, but no requests-per-second, caching, CDN or quantitative performance target is documented.	Flow Designer / Service Catalog; quantitative design not specified

Documented Implementation Flow

Milestone 1 — Flow

Create a Flow named "Standard laptop task", use the Global application and System user execution context, add a Service Catalog trigger, then add the Create Catalog Task action. The action uses the Requested Item Record, sets the short description and description to "Laptop need to Configured", assigns the Hardware group, sets Approval to Approved, saves and activates the flow.

Milestone 2 — Flow Assignment

Open Maintain Items, locate the Standard Laptop item, remove the remaining automations, add the created "Standard Laptop Task" flow, and save the record.

Milestone 3 — Service Catalog

Use Service Catalog → Hardware → Standard Laptop → Order Now. Review the order status, open the Request Number, review Approvers, approve the request, open the Requested Item, then review the Catalog Tasks section for the updated status, short description and assignment group.

Technology-to-Requirement Traceability

Requirement / Need	ServiceNow implementation element	Expected outcome documented in source
Reduce manual procurement work	Flow Designer automation	Reduced manual intervention and potential errors
Ensure configuration task is generated	Create Catalog Task action	Task is automatically created after the documented trigger/approval flow
Route work to hardware team	Assignment group = Hardware	Task is assigned to the Hardware group
Standardize task information	Short description + Description values	Catalog task contains "Laptop need to Configured"
Respect approval state	Approval field = Approved	Catalog task action is configured with Approved
Connect automation to catalog item	Maintain Items → Standard Laptop → Process engine / Flow	Standard Laptop service item uses the created flow

Conclusion

By automating the standard laptop procurement process with ServiceNow Flow Designer, the documented solution addresses procurement inefficiencies and manual overhead. The workflow is intended to support timely laptop configuration, reduce user wait time, optimize resource allocation, and improve the user experience.

Source-bound note: The project document does not specify a separate database, cloud database, file-storage service, external API, machine-learning model, detailed security controls, scalability architecture, availability design, or quantitative performance targets. Those fields have therefore been retained as "Not specified" rather than inferred.

References / Source Materials

1. Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer — supplied project documentation.
2. Technology Stack (Architecture & Stack) — supplied template used as the structural reference for this deliverable.